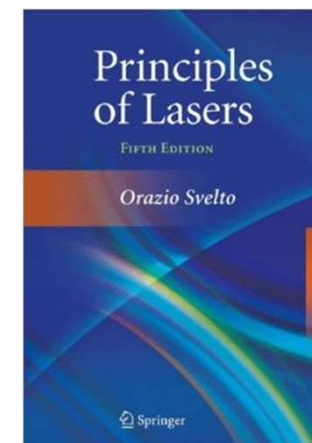




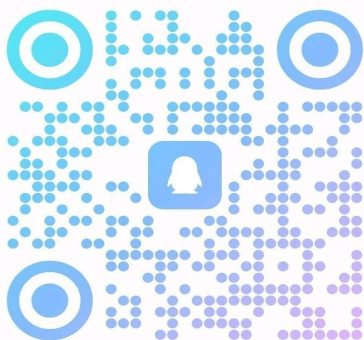
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers



2026春季 激光原理与...
群号: 1078576936



Preface General Introduction

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Email: luod@sustech.edu.cn

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of Engineering South Tower

Tel: 0755-88018552



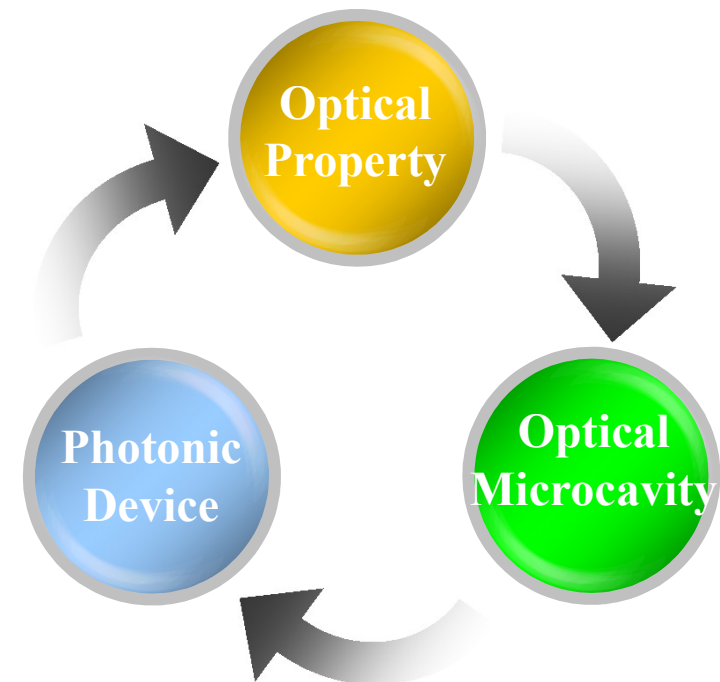
Laser Spectroscopy



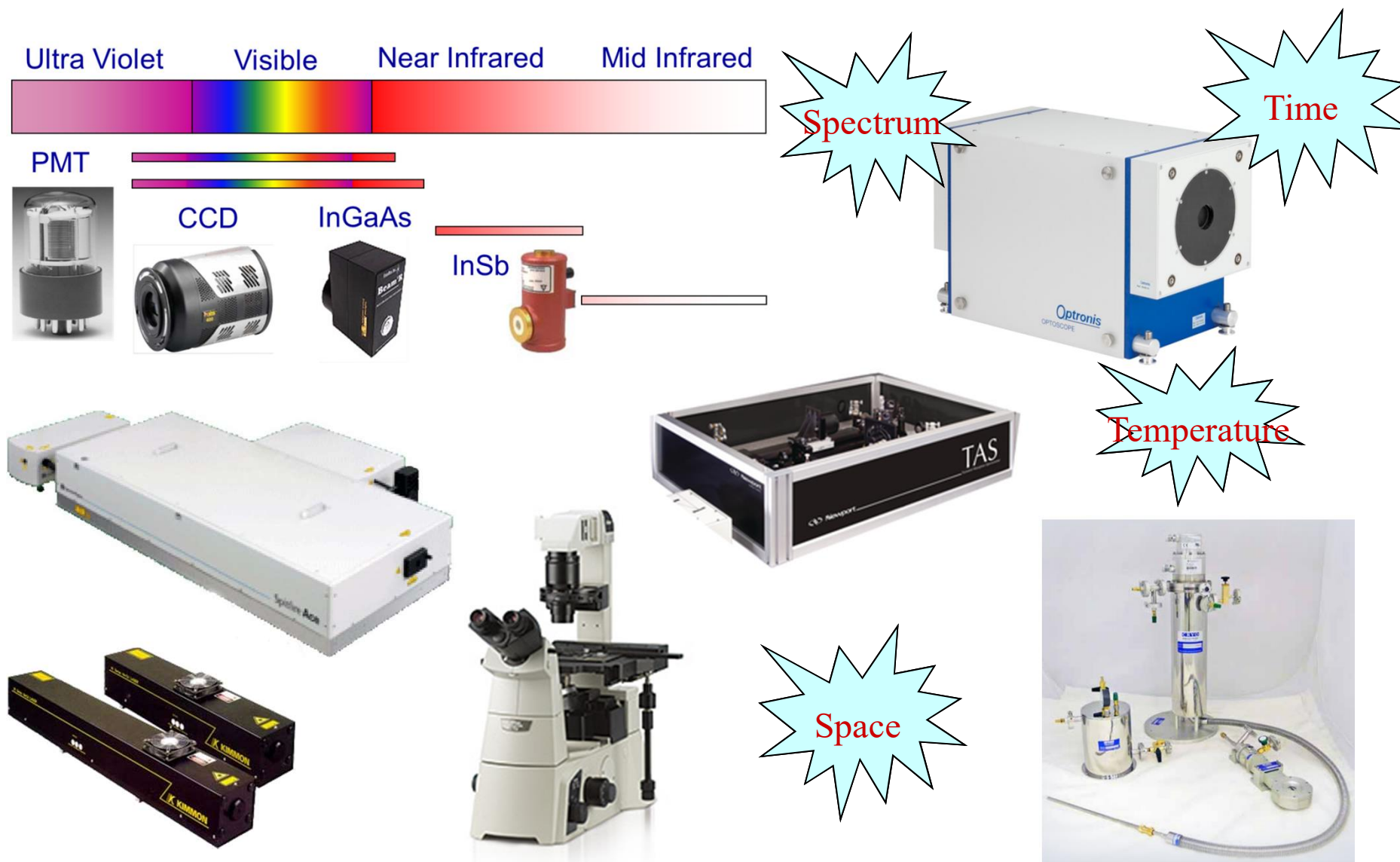
- ✓ **Steady State Spectroscopy** (*Absorption, Reflection, Photoluminescence, Photoluminescence Excitation, etc*)
- ✓ **Nonlinear Optics** (*Z-Scan, Multi-Photon Spectroscopy*)
- ✓ **Time-Resolved Spectroscopy** (*Time-Resolved Photoluminescence*)
- ✓ **Ultra-Fast Pump-Probe Transient Absorption Spectroscopy**

Research Summarize

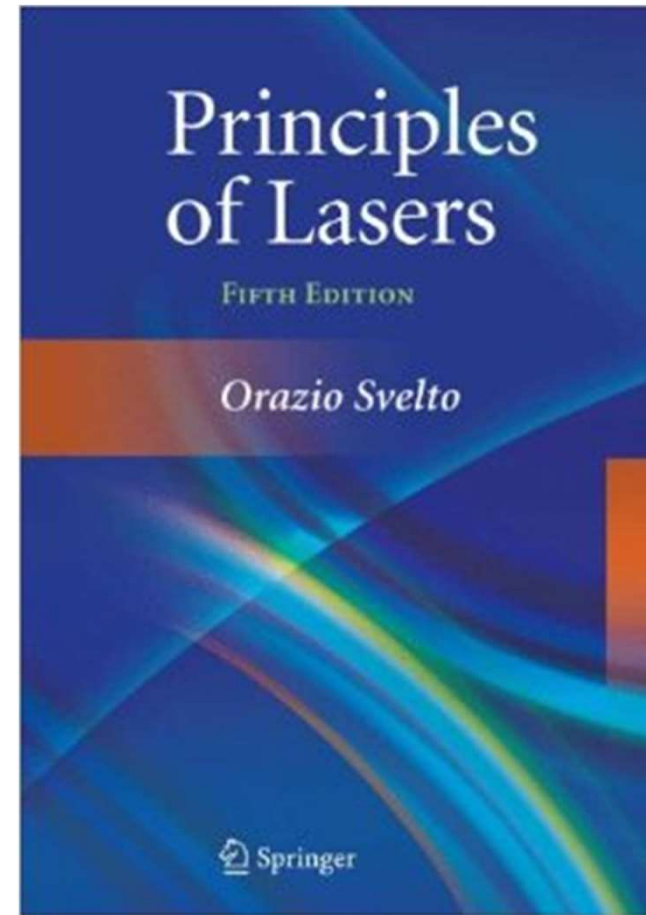
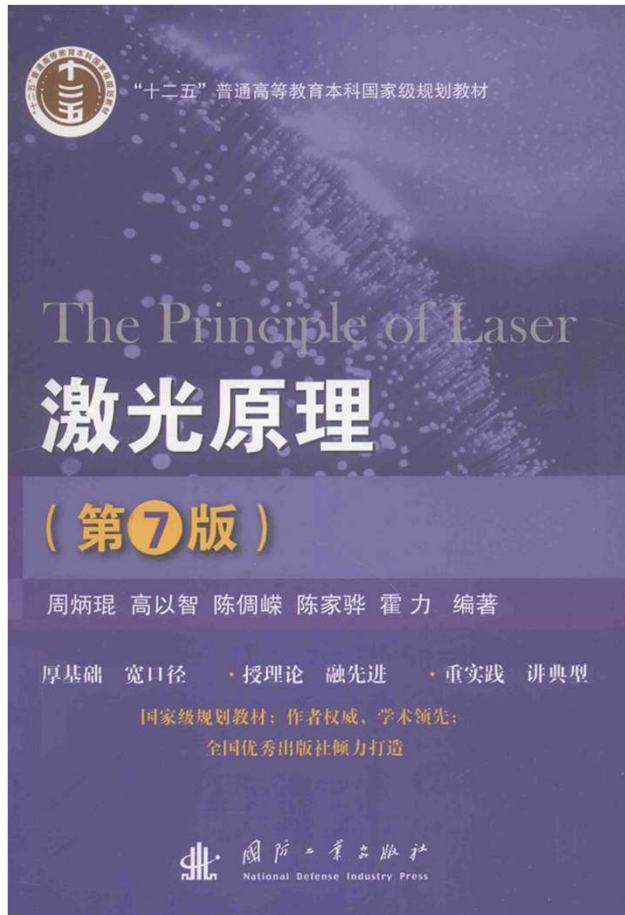
Optical Property	Investigation of Carrier Dynamic; Excitonic Property; Energy Transfer
Optical Microcavity	Fabry-Pérot Cavity; Random Cavity; Whispering Gallery Modes
Photonic Device	Light Emitting Diode; Temperature Sensor; Refractive Index Sensor



Laser Spectroscopy



Course References



Principles and Technologies of Lasers

Course Information

Course ID: EE310

Name: Principles and Technologies of Lasers

Credit: 3

Content:

Section 1: Basic principle

➤ Chapter **1**

Section 2: Optical cavity oscillation

➤ Chapter **2**

Section 3: Laser oscillation & amplification

➤ Chapter **3, 4, 5**

Section 4: Control & improvement technique

➤ Chapter **6**

Section 5: Typical lasers and light amplifier

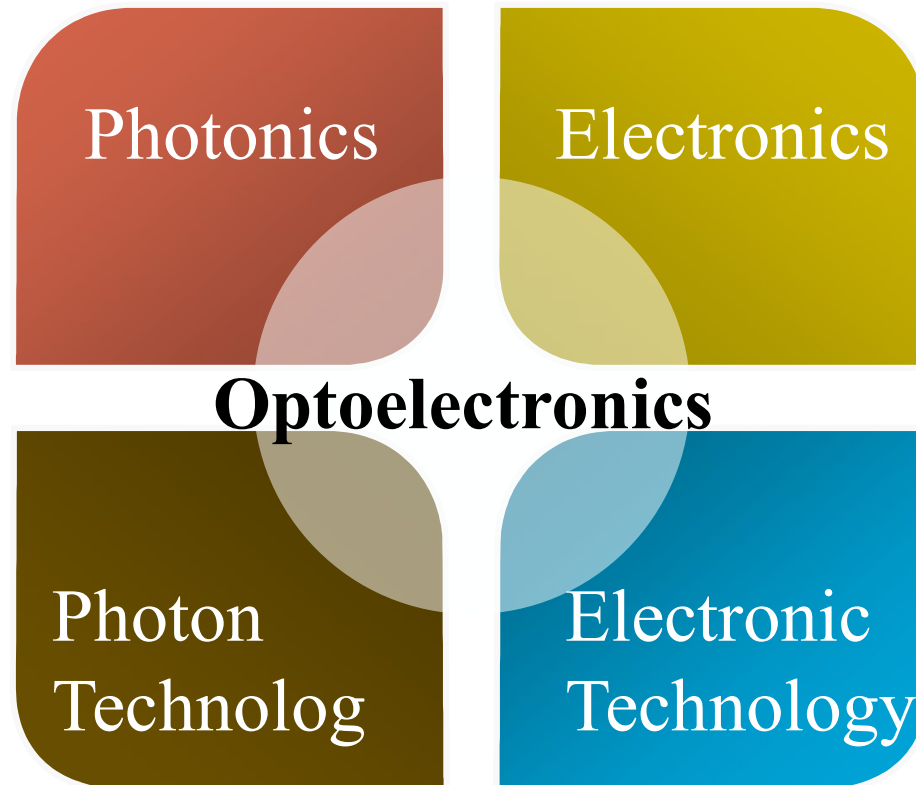
➤ Chapter **7, 8**

Final Evaluation

- 1. Attendance 10%**
- 2. Homework 20%**
- 3. Mid-term exams 30%**
- 4. Final exam 40%**

Photon as a Carrier for
Information and Energy

Electron as a Carrier for
Information and Energy



The Coherent of Light

- **Generation: Lasers**
- Modulation
- Detection & Application

Electronic Components

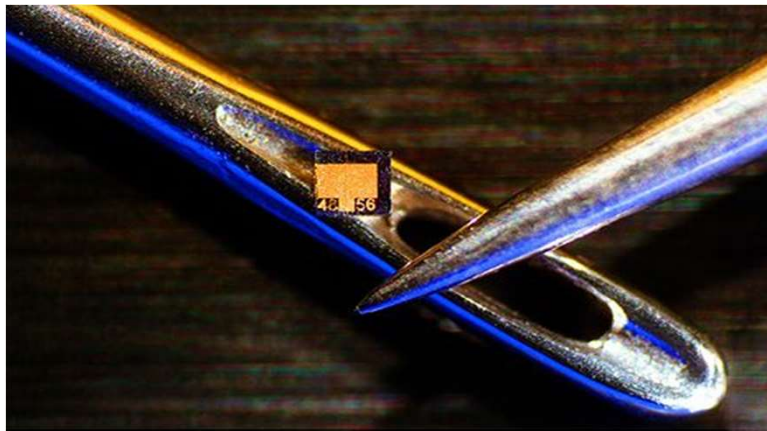
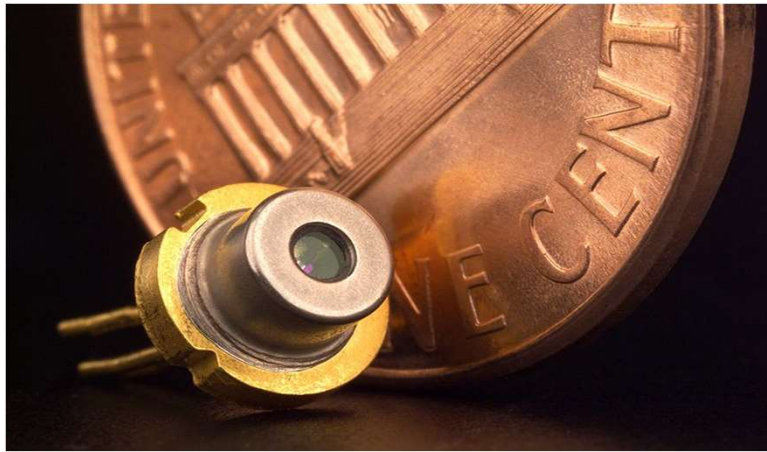
- Functional Circuit Design and Manufacturing
- Including Information Technology and Power Electronics

Why the interest of Lasers

- ❖ Lasers have unique properties
- ❖ Created many new devices
- ❖ Improved existing devices

Examples of Laser application

- ❖ Bar-code readers, sensors
- ❖ Compact discs, computer printers
- ❖ Laser show, holography – 3D
- ❖ Position & motion control
- ❖ Non-destructive spectroscopy measurements
- ❖ Military system, medical procedures
- ❖ Fiberoptic communication



Lasers range in size from microscopic diode lasers (top) with numerous applications, to football field sized neodymium glass lasers (bottom) used for inertial confinement fusion, nuclear weapons research and other high energy density physics experiments.

Warning symbol
for lasers



Military Application

laser designator



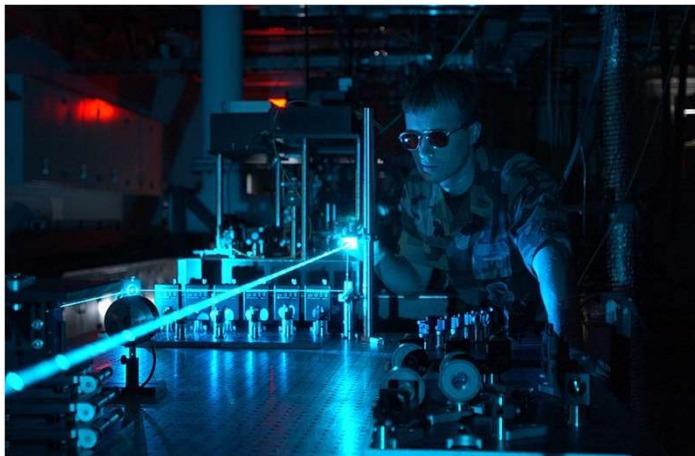
Laser Sight



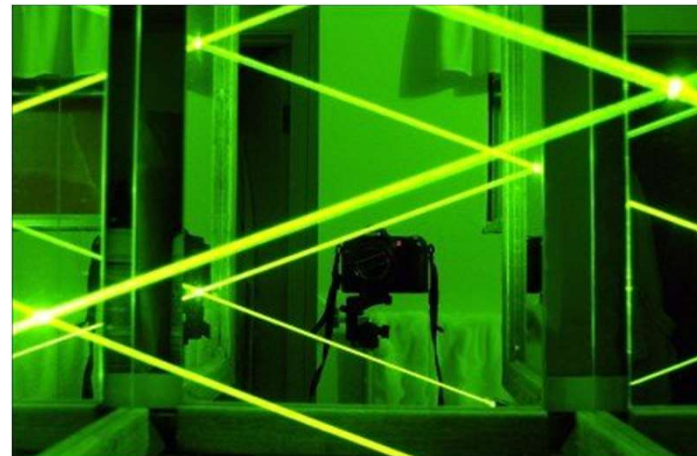
Anti-Satellite Laser



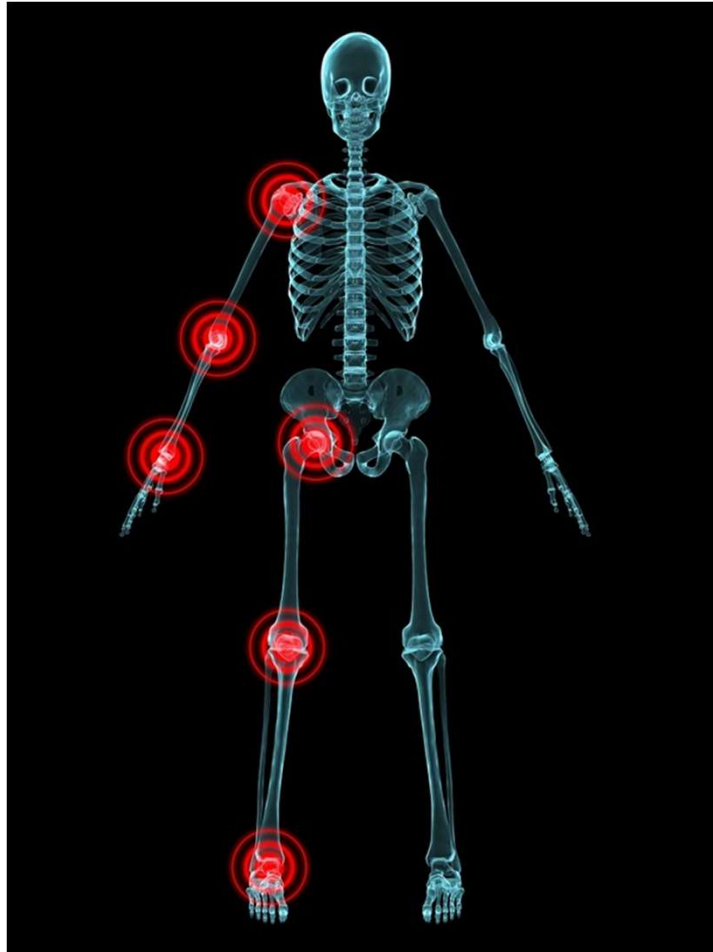
Military laser



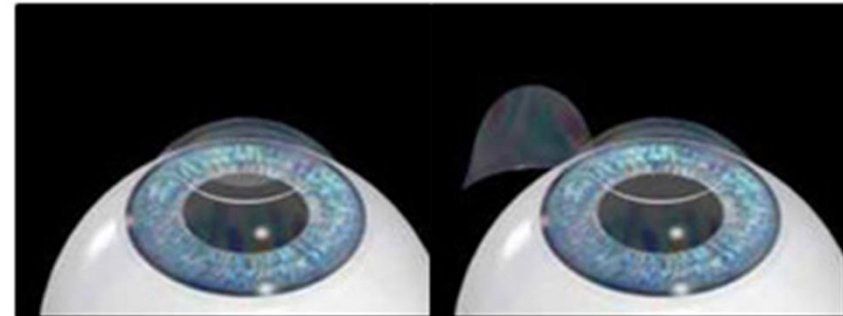
Safety



Medical Application

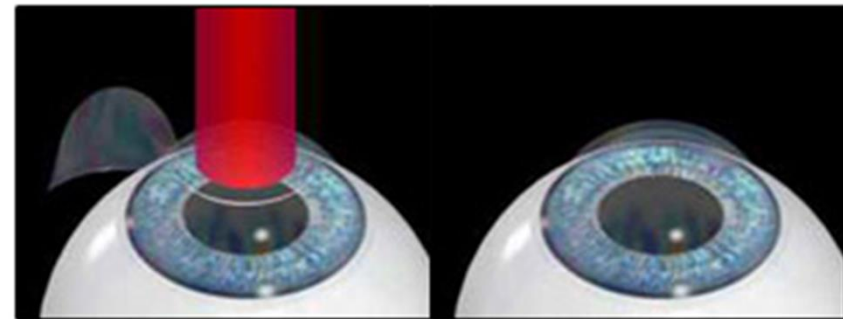


Low-level laser therapy



Step 1: Corneal flap is created with a microkeratome.

Step 2: The corneal flap is folded back.



Step 3: Excimer laser beam reshapes the cornea.

Step 4: The corneal flap is folded back in place.

Laser eye surgery

Industrial and commercial



Laser pointer



Laser line level



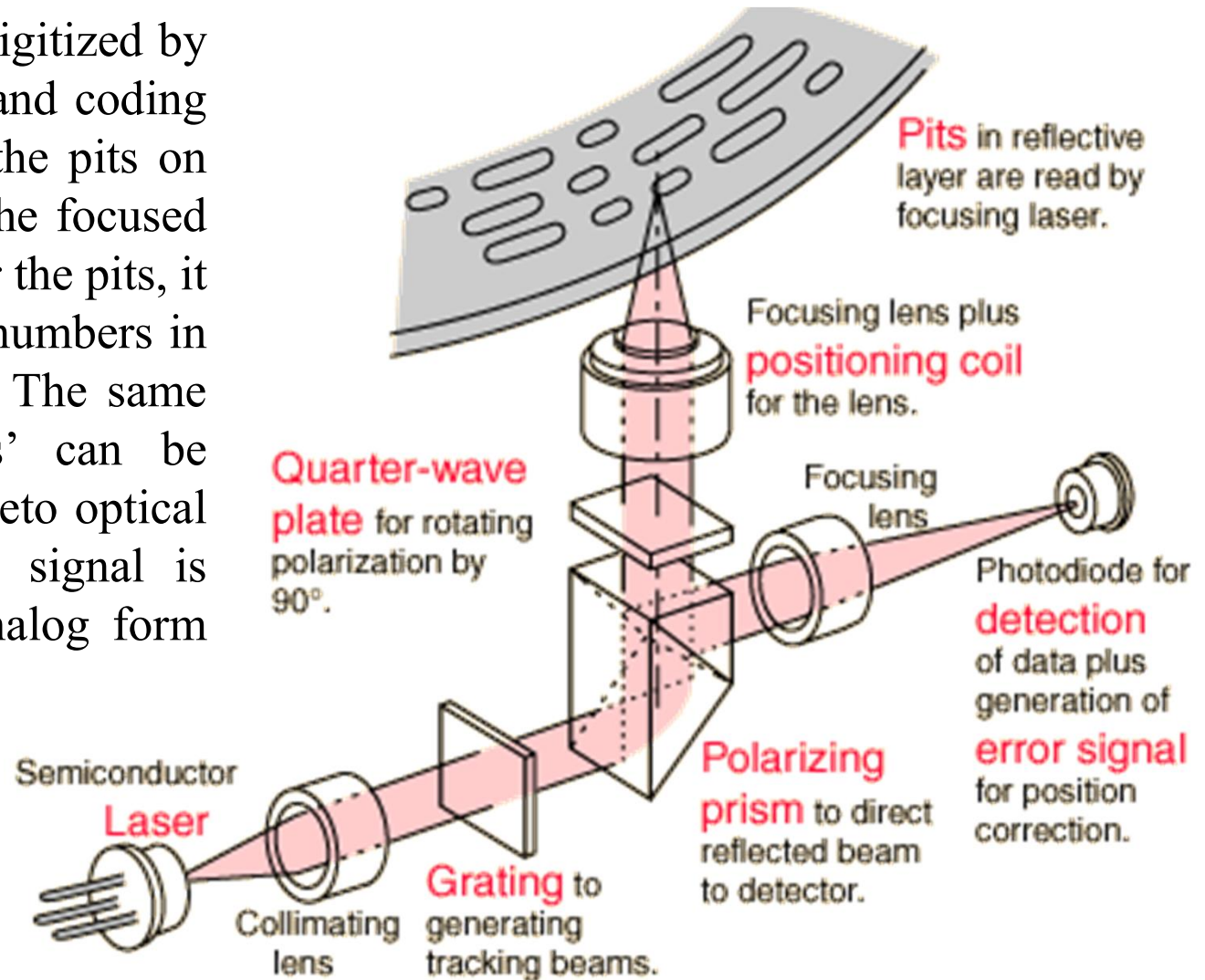
Laser light display

3D laser scan



Compact Disc Audio

Analog sound data is digitized by sampling at 44.1 kHz and coding as binary numbers in the pits on the compact disc. As the focused laser beam sweeps over the pits, it reproduces the binary numbers in the detection circuitry. The same function as the 'pits' can be accomplished by magneto optical recording. The digital signal is then reconverted to analog form by a D/A converter.



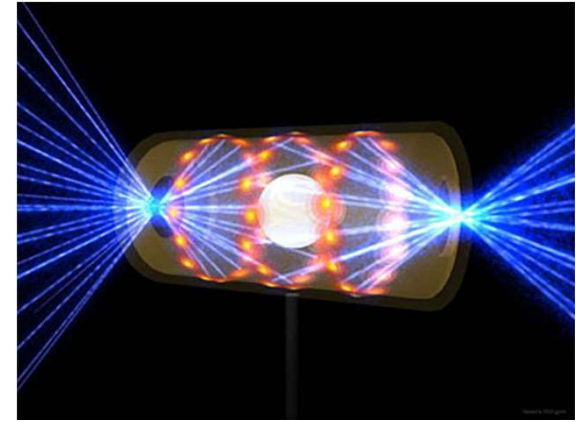
Other Laser Applications



Micro-machining



Laser derusting



Laser induced fusion

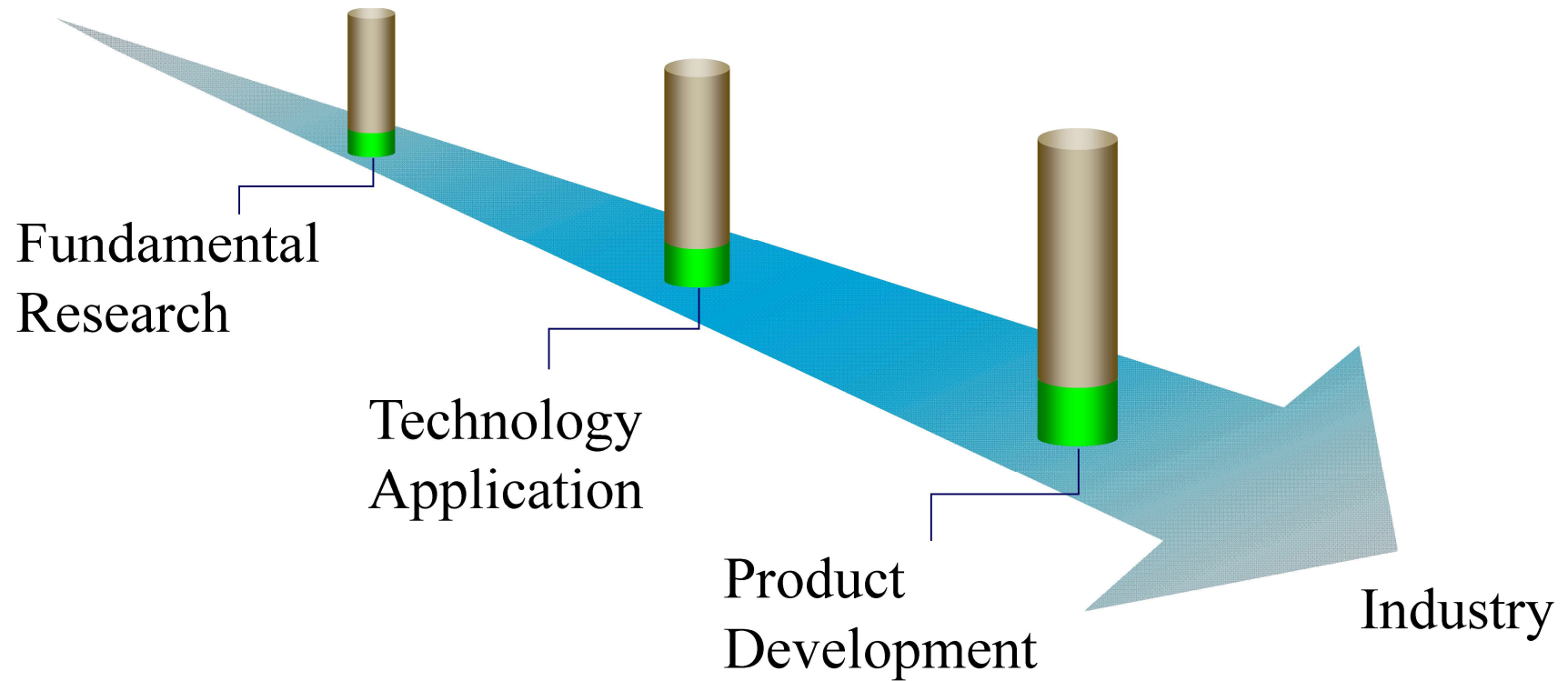


Laser weapon



Mosquito eradication

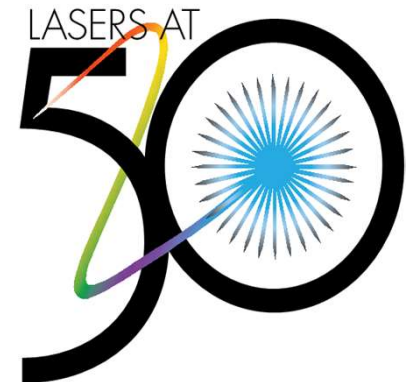
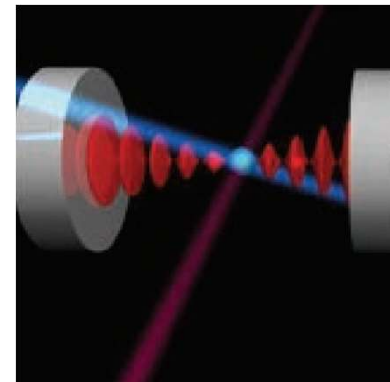
Development of Technology



Historical Evolution of Laser



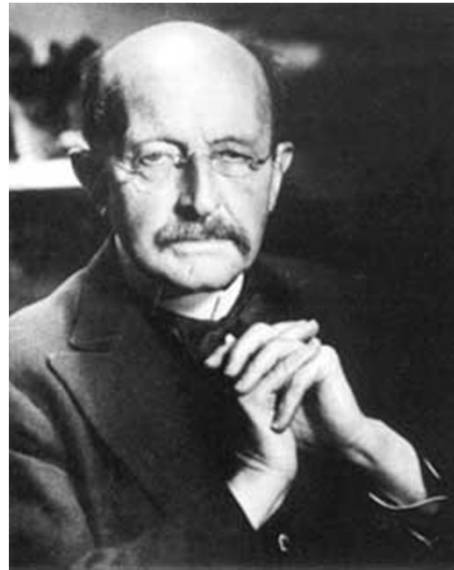
- ❖ Since its invention half a century ago, the laser has made possible a staggering number of applications in a wide range of fields: industry, science, medicine, the military and more.
- ❖ This section discusses the historical evolution of laser, and the key players who influenced the development of this important technology.



1905 Photon

- Some theorists were on the right track, especially Plank, who proposed that nature acted by using "quanta" of energy. But it was the young, unknown Albert Einstein who explained everything and started the field of quantum mechanics with his paper on the photoelectric effect (1905).

Einstein showed that light does not consist of continuous waves, nor of small, hard particles. Instead, it exists as bundles of wave energy called photons.



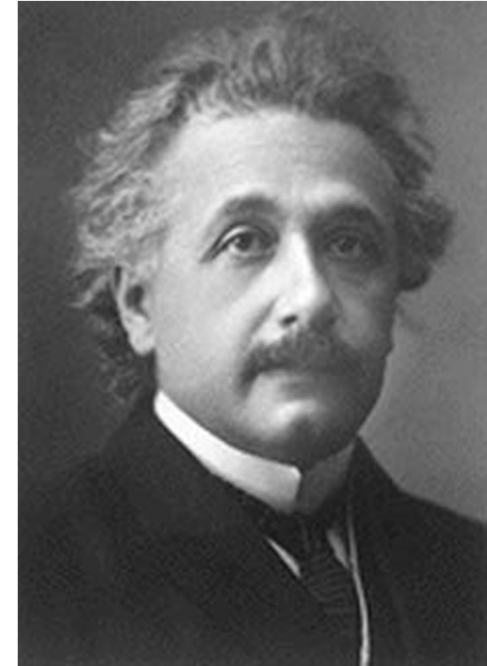
Plank



Albert Einstein

1917 Stimulated Emission

- Einstein proposed the process that makes lasers possible, called **stimulated emission**. He theorized that, besides absorbing and emitting light spontaneously, electrons could be stimulated to emit light of a particular wavelength
- But it would take nearly 40 years before scientists would be able to amplify those emissions, proving Einstein correct and putting lasers on the path to becoming the powerful and ubiquitous tools they are today.



"A splendid light has dawned on me..."

- *Albert Einstein*

Microwave Cavity and Accelerator

UNITED STATES PATENT OFFICE

2,190,712

HIGH EFFICIENCY RESONANT CIRCUIT

William W. Hansen, Stanford University, Calif.,
assignor to The Board of Trustees of The
Leland Stanford Junior University, Stanford
University, Calif., a corporation of California

Application July 27, 1936, Serial No. 92,757

Feb. 20, 1940.

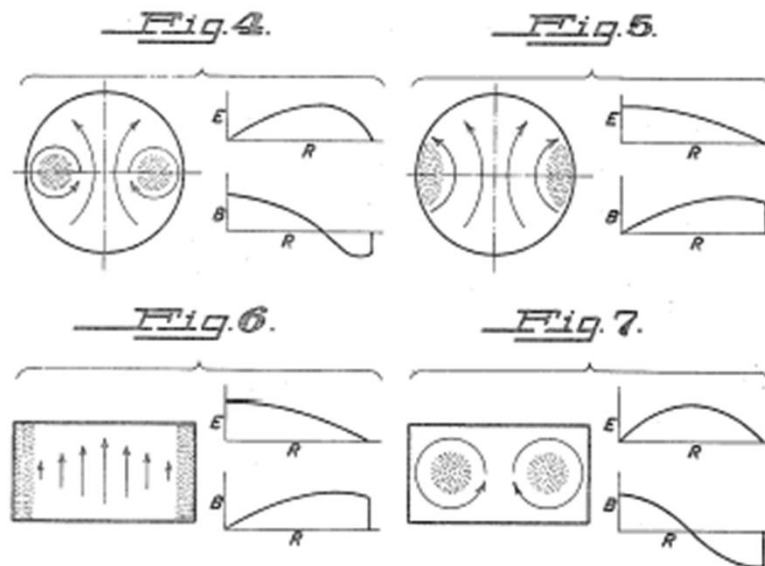
W. W. HANSEN

2,190,712

HIGH EFFICIENCY RESONANT CIRCUIT

Filed July 27, 1936

3 Sheets-Sheet 2



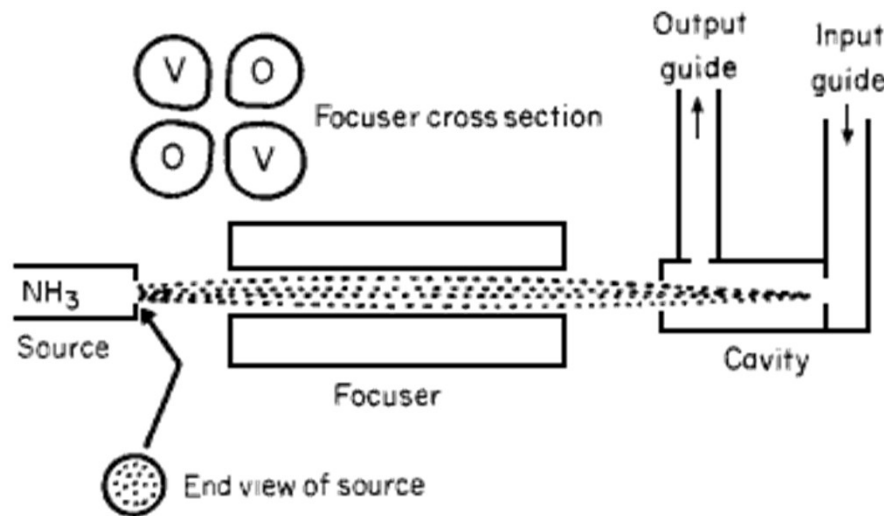
Invented and patented the
microwave cavity (1936)



The linear accelerator (1947)

1954 First Microwave Laser

- The first population inversion was achieved in the ammonia molecule, which consists of a nitrogen at the apex of a pyramid of three hydrogen atoms. The two lowest levels of ammonia are the result of inversion splitting of the vibrational levels caused by a potential curve with a double minima. Population inversion in ammonia is established by physical separation of molecules in the upper quantum state from those in the lower quantum state.



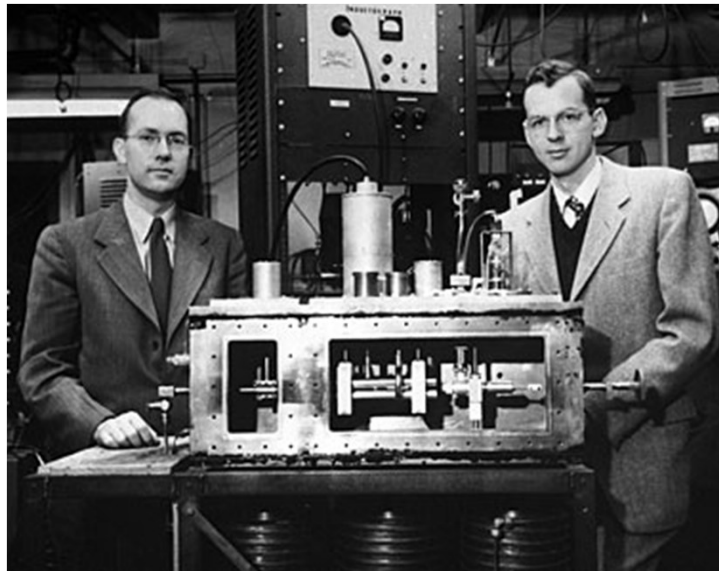
Beam of ammonia passes through an electrostatic focuser to separate out molecules in the upper quantum state. (Townes, 1954)

1954 First Microwave Laser

- Working with Herbert J. Zeiger and graduate student James P. Gordon, Townes demonstrates the first maser at Columbia University.
- The ammonia **M**icrowave **A**mplification by **S**timulated **E**mission of **R**adiation (**maser**), the first device based on Einstein's predictions.
- The maser radiates at a wavelength of a little more than 1 cm and generates approximately 10 nW of power.



Charles Hard Townes



Townes & James P. Gordon



Herbert J. Zeiger

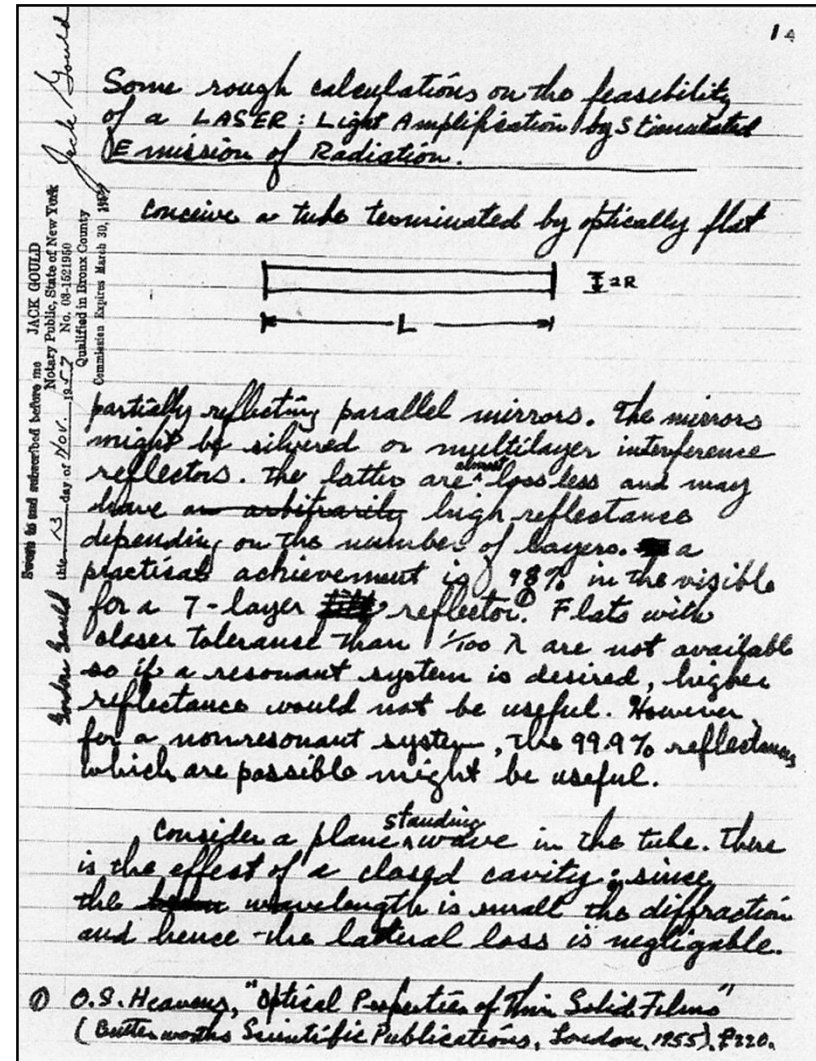
- **1955:** At P.N. Lebedev Physical Institute in Moscow, Nikolai G. Basov and Alexan-der M. Prokhorov attempt to design and build oscillators. They propose a method for the production of a negative absorption that was called the pumping method.
- **1956:** Nicolaas Bloembergen of Harvard University develops the microwave solid-state maser.
- **Sep. 14, 1957:** Townes sketches an early optical maser in his lab notebook.
- **1958:** Townes, a consultant for Bell Labs, and his brother-in-law, Bell Labs researcher Arthur L. Schawlow, in a joint paper published in *Physical Review Letters*, show that masers could be made to operate in the optical and infrared regions and propose how it could be accomplished. At Lebedev Institute, Basov and Prokhorov also are exploring the possibilities of applying maser principles in the optical region.



Arthur L. Schawlow

Patent Dispute of the First Conceive of Laser

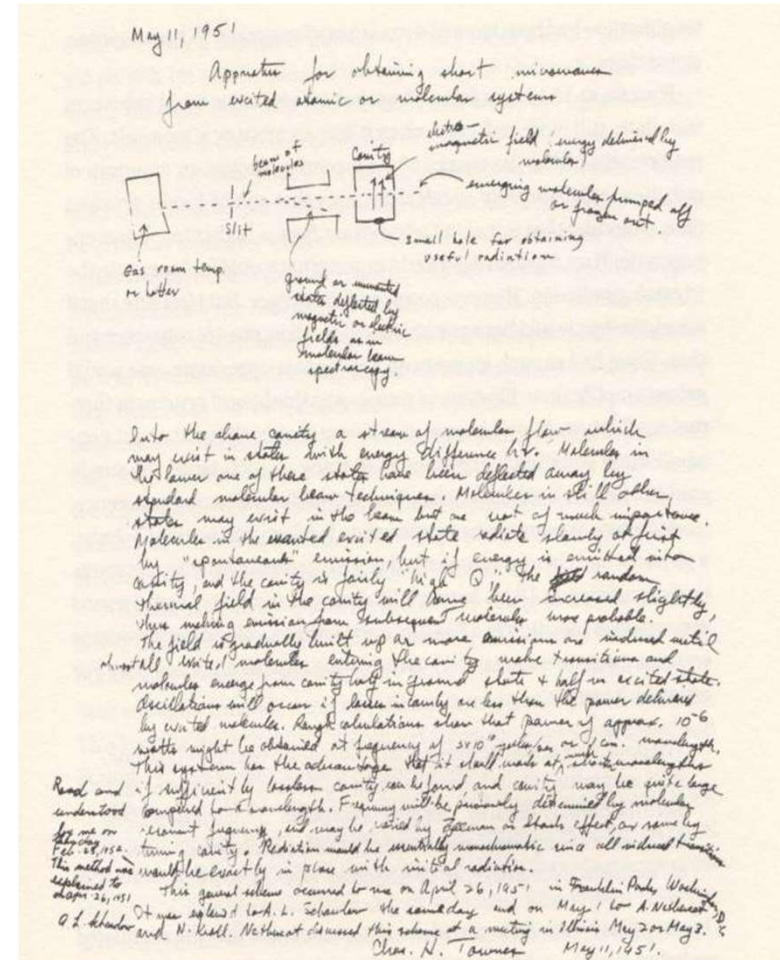
- Columbia University graduate student Gordon Gould jots his ideas for building a laser in his notebook and has it notarized at a candy store in the Bronx. It is considered the first use of the acronym laser. Gould leaves the university a few months later to join private research company TRG (Technical Research Group), dated **Nov. 13, 1957**.
- **April 1959**: Gould and TRG apply for laser-related patents stemming from Gould's ideas.



Patent Dispute of the First Conceive of Laser



- **March 22, 1960:** Townes and Schawlow, under Bell Labs, are granted US patent number 2,929,922 for the optical maser, now called a laser. With their application denied, Gould and TRG launch what would become a 30-year patent dispute related to laser invention.





Nobel laureate and laser inventor Charles Townes dies at 99

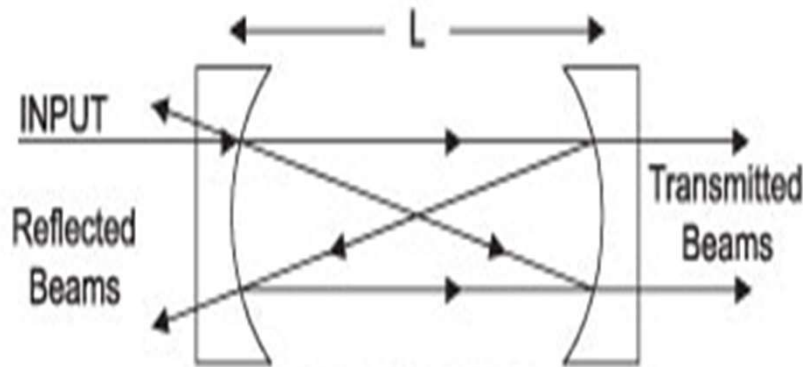
By Robert Sanders, Media Relations | January 27, 2015

BERKELEY — Charles Hard Townes, a professor emeritus of physics at the University of California, Berkeley, who shared the 1964 Nobel Prize in Physics for invention of the laser and subsequently pioneered the use of lasers in astronomy, died early Tuesday, Jan. 27, in Oakland. He was 99 and in failing health, and died on his way to the hospital.

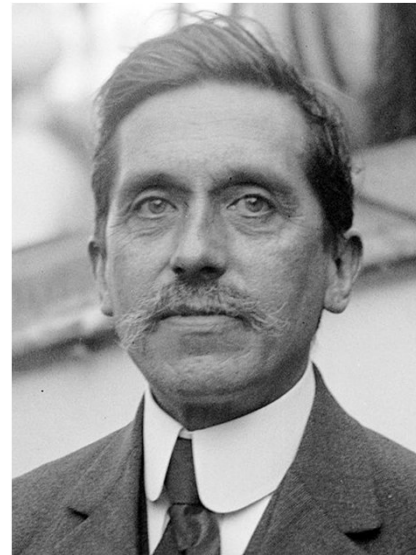
1960 First Optical Laser

❖ Fabry-Pérot Cavity

- Two perfectly parallel mirrors were the key to stimulating both solid and gaseous molecules to produce an inverted population. Without the Fabry-Pérot device, you're just exciting particles randomly and to no avail.



19th century



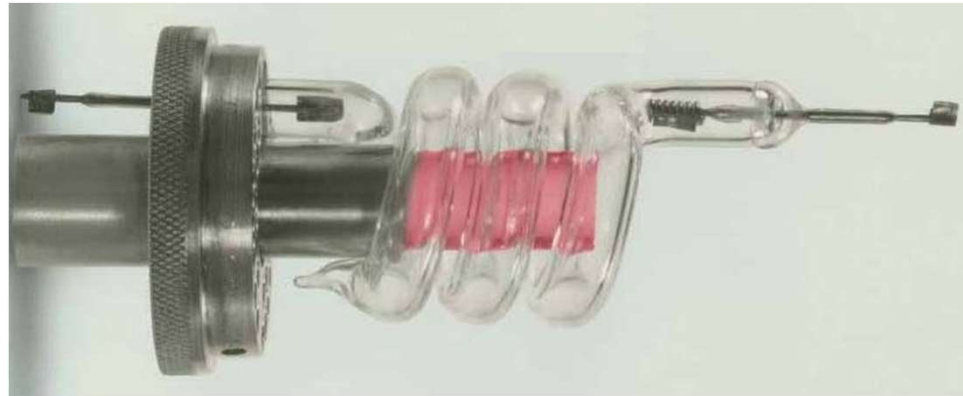
Charles Fabry



Alfred Pérot

1960 First Optical Laser

Theodore H. Maiman, a physicist at Hughes Research Laboratories in Malibu, Calif., constructs the first laser using a cylinder of synthetic ruby measuring 1 cm in diameter and 2 cm long, with the ends silver-coated to make them reflective and able to serve as a Fabry-Pérot resonator. Maiman uses photographic flashlamps as the laser's pump source.



Paper submitted for publication **Rejected**

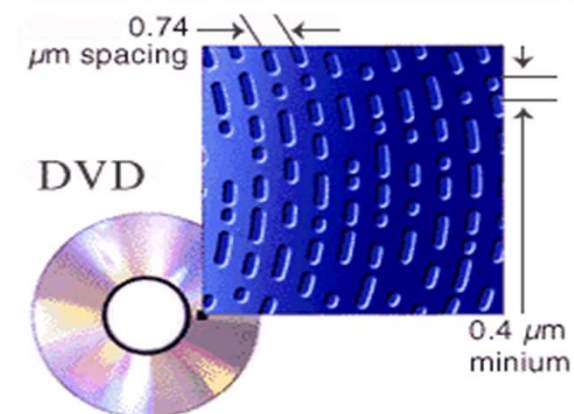
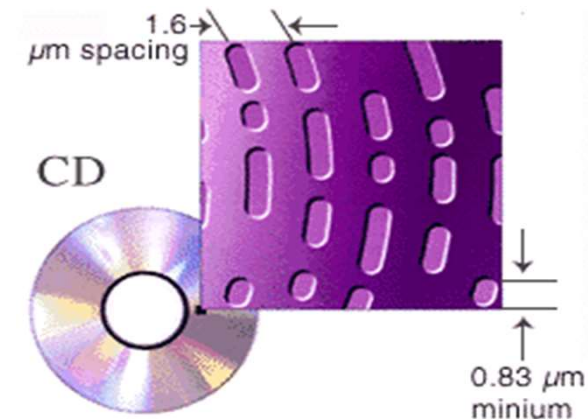
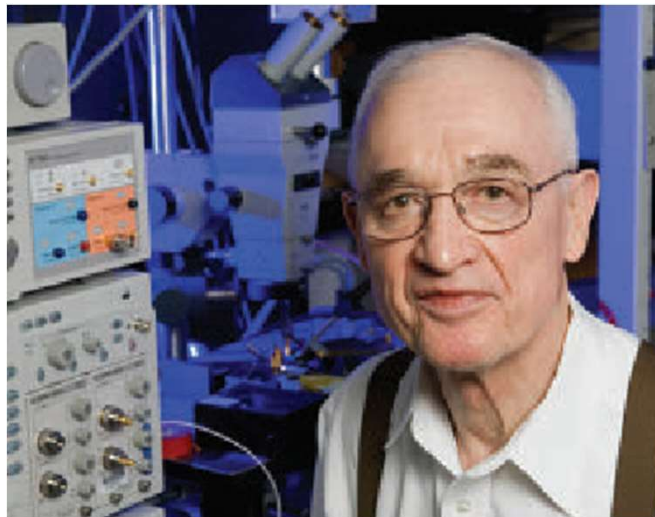
Results announced in New York Times, 8 July 1960.

Paper accepted by “Nature”, appeared 6 August 1960.

- **December 1960:** Ali Javan, William Bennett Jr. and Donald Herriott of Bell Labs develop the helium-neon (HeNe) laser, the first to generate a continuous beam of light at 1.15 μm .
- **1961:** Lasers begin appearing on the commercial market through companies such as Trion Instruments Inc., Perkin-Elmer and Spectra-Physics.
- **October 1961:** American Optical Co.'s Elias Snitzer reports the first operation of a neodymium glass (Nd:glass) laser.
- **December 1961:** The first medical treatment using a laser on a human patient is performed by Dr. Charles J. Campbell of the Institute of Ophthalmology at Columbia-Presbyterian Medical Center and Charles J. Koester of the American Optical Co. at Columbia-Presbyterian Hospital in Manhattan. An American Optical ruby laser is used to destroy a retinal tumor.

❖ **October 1962:** Nick Holonyak Jr., a consulting scientist at a General Electric Co. lab in Syracuse, N.Y., publishes his work on the “visible red” GaAsP laser diode, a compact, efficient source of visible coherent light that is the basis for today’s red LEDs used in consumer products such as CDs, DVD players and cell phones.

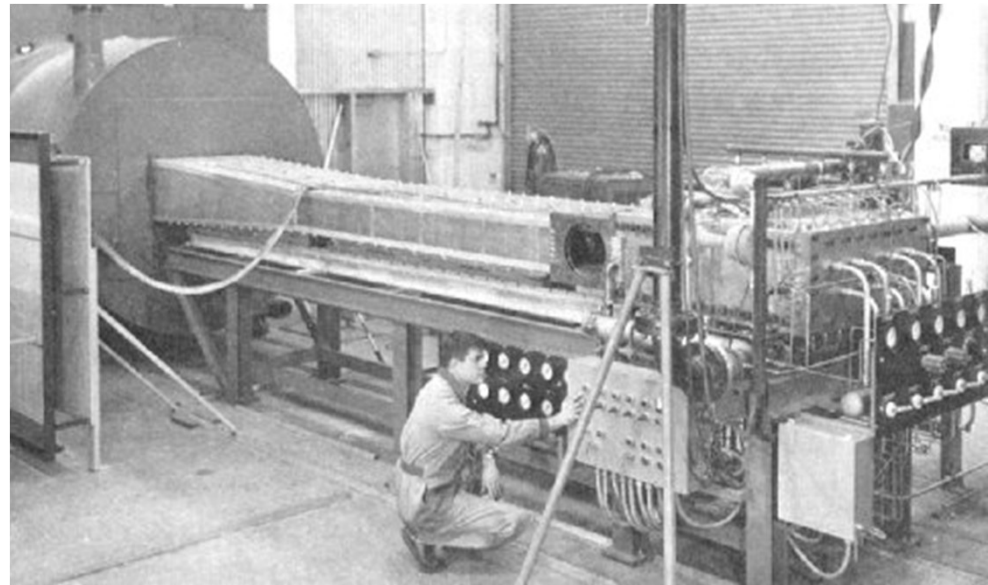
- He also invented the first visible LED in 1962 while working as a consulting scientist at a General Electric Company laboratory in Syracuse, New York and has been called "the father of the light-emitting diode"



1964 First Gas Dynamic Laser

- ❖ The carbon dioxide laser is invented by Kumar Patel at Bell Labs. The most powerful continuously operating laser of its time, it is now used worldwide as a cutting tool in surgery and industry.

Large scale 135 Kilowatt gasdynamic laser at Avco Everett Research Lab, Inc. was among the first very high power lasers. Initially this research was classified by the U.S. government, even today information on these types of lasers is scarce. (Gerry, 1970)



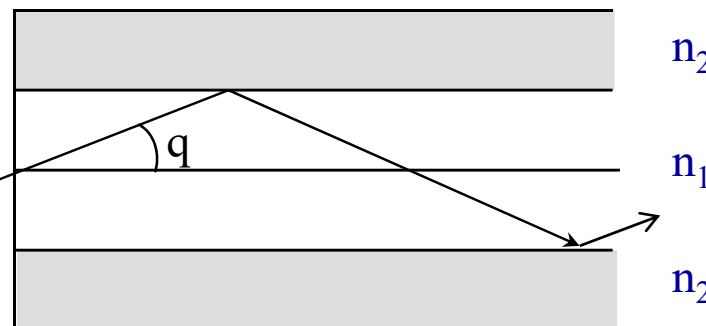
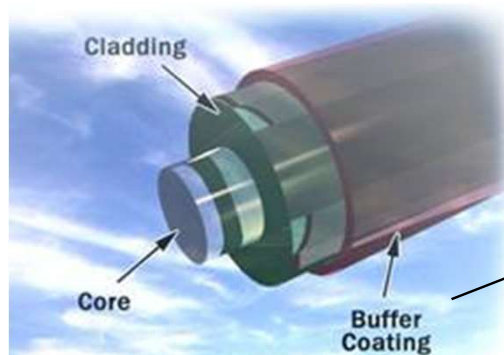
- ❖ **1964:** The Nd:YAG (neodymium-doped YAG) laser is invented by Joseph E. Geusic and Richard G. Smith at Bell Labs. The laser later proves ideal for cosmetic applications, such as laser-assisted in situ keratomileusis (lasik) vision correction and skin resurfacing.
- ❖ **1964:** Townes, Basov and Prokhorov are awarded the Nobel Prize in physics for their “fundamental work in the field of quantum electronics, which has led to the construction of oscillators and amplifiers based on the maser-laser-principle.”
- ❖ **March 1964:** After working for two years on HeNe and xenon lasers, William B. Bridges of Hughes Research Labs discovers the pulsed argon-ion laser, which, although bulky and inefficient, could produce output at several visible and UV wavelengths.

1966 Break through of fiber optics

Charles K. Kao, working with George Hockham at Standard Telecommunication Laboratories in Harlow, England, makes a discovery that leads to a breakthrough in fiber optics. He calculates how to transmit light over long distances via optical glass fibers, deciding that, with a fiber of purest glass, it would be possible to transmit light signals over a distance of 100 km, compared with only 20 m for the fibers available in the 1960s. Kao receives a 2009 Nobel Prize in physics for his work.



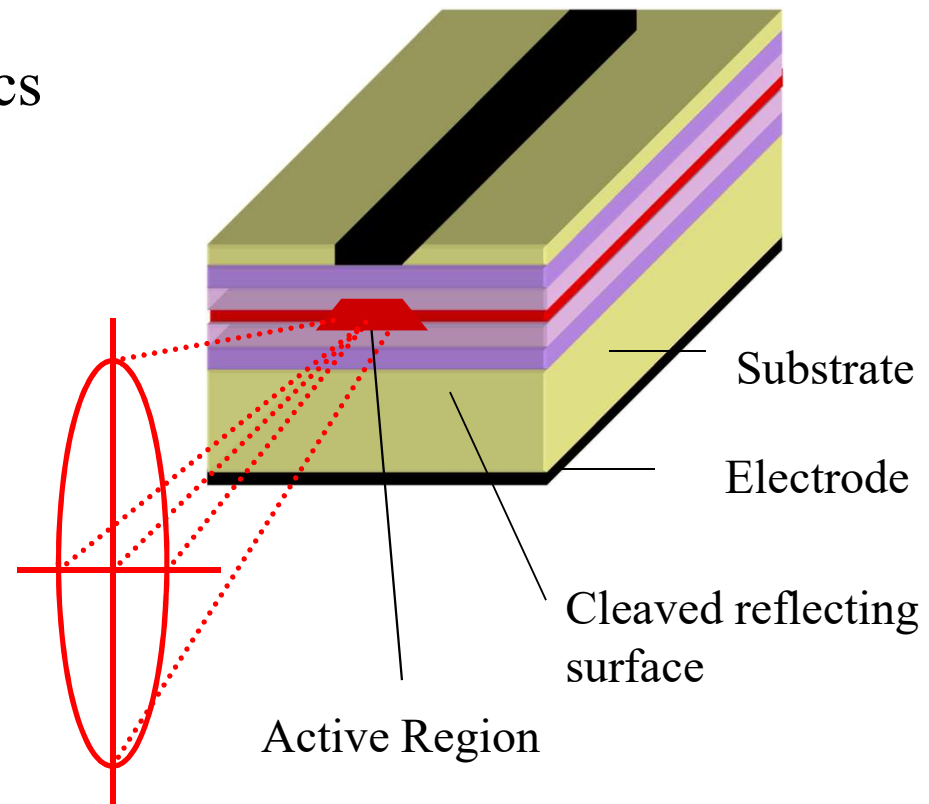
04-Nov-1933 to 23-Sep-2018



1970 First Semiconductor Laser

- ❖ **Zhores** Alferov's group at the Ioffe Physico-Technical Institute in Russia and Mort Panish and Izuo Hayashi at Bell Labs produce the first CW RT semiconductor lasers, paving the way toward commercialization of fiber optics communications.

2000 Nobel Laureate in Physics



- **1975:** Engineers at Laser Diode Labs Inc. develop the first commercial continuous-wave semiconductor laser operating at room temperature. Continuous-wave operation enables the transmission of telephone conversations.
- **1978:** Following the failure of its videodisc technology, Philips announces the compact disc (CD) project.
- **1978:** The LaserDisc hits the home video market, with little impact. The earliest players use HeNe laser tubes to read the media, while later players used infrared laser diodes.
- **1982:** Peter F. Moulton of MIT's Lincoln Laboratory develops the titanium-sapphire laser, used to generate short pulses in the picosecond and femtosecond ranges. The Ti:Sapphire laser replaces the dye laser for tunable and ultrafast laser applications.

- **1994:** The first demonstration of a quantum dot laser with high threshold density was reported by Nikolai N. Ledentsov of A.F. Ioffe Physico-Technical Institute in Leningrad.
- **1994:** The first semiconductor laser that can simultaneously emit light at multiple widely separated wavelengths — the quantum cascade (QC) laser — is invented at Bell Labs by Jerome Faist, Federico Capasso, Deborah L. Sivco, Carlo Sirtori, Albert L. Hutchinson and Alfred Y. Cho. The laser is unique in that its entire structure is manufactured a layer of atoms at a time by the crystal growth technique called molecular beam epitaxy. Simply changing the thickness of the semiconductor layers can change the laser's wavelength. With its room-temperature operation and power and tuning ranges, the QC laser ideal for remote sensing of gases in the atmosphere.

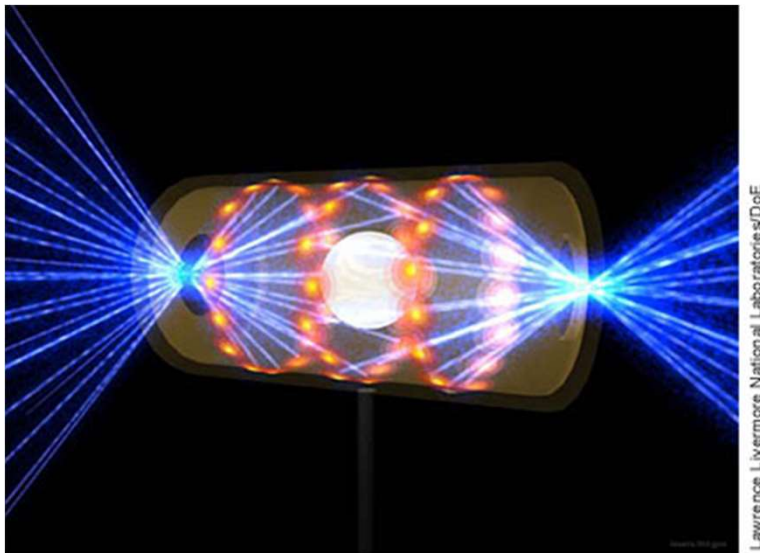
- **January 1997:** Shuji Nakamura, Steven P. DenBaars and James S. Speck at the University of California, Santa Barbara, announce the development of a gallium-nitride (**GaN**) laser that emits bright blue-violet light in pulsed operation.
- **2004:** Electronic switching in a **Raman laser** is demonstrated for the first time by Ozdal Boyraz and Bahram Jalali of the University of California, Los Angeles. The first silicon Raman laser operates at room temperature with 2.5-W peak output power. In contrast to traditional Raman lasers, the pure-silicon Raman laser can be directly modulated to transmit data.

- **September 2006:** John Bowers and colleagues at the University of California, Santa Barbara, and Mario Paniccia, director of Intel Corp.'s Photonics Technology Lab in Santa Clara, Calif., announce that they have built the first electrically powered **hybrid silicon laser** using standard silicon manufacturing processes.
- **August 2007:** Bowers and his doctoral student Brian Koch announce that they have built the first mode-locked **silicon evanescent laser**, providing a new way to integrate optical and electronic functions on a single chip and enabling new types of integrated circuits.
- **May 29, 2009:** The largest and highest-energy laser in the world, the National Ignition Facility (NIF) at Lawrence Livermore National Laboratory in Livermore, Calif., is dedicated. In a few weeks, the system begins firing all 192 of its laser beams onto targets.

June 2009: NASA launches the Lunar Reconnaissance Orbiter (LRO). LOLA, the Lunar Orbiter Laser Altimeter on the LRO, will use a laser to gather data about the high and low points on the moon. NASA will use that information to **create 3-D maps** that could help determine lunar ice locations and safe landing sites for future spacecraft.



While orbiting the moon, the Lunar Reconnaissance Orbiter will take pictures and gather information about the moon's surface.



Artist's rendition showing a National Ignition Facility target pellet inside a hohlraum capsule with laser beams entering through openings on either end. The beams compress and heat the target to the necessary conditions for nuclear fusion to occur.

January 2010: The National Nuclear Security Administration announces that NIF has successfully delivered a historic level of laser energy — more than 1 MJ — to a target in a few billionths of a second and demonstrated the target drive conditions required to achieve **fusion ignition**, a project scheduled for the summer of 2010. The peak power of the laser light is about 500 times that used by the US at any given time.

The Nobel Prize in Physics 2018



Ill. Niklas Elmehed. © Nobel Media

Arthur Ashkin

Prize share: 1/2



Ill. Niklas Elmehed. © Nobel Media

Gérard Mourou

Prize share: 1/4



Ill. Niklas Elmehed. © Nobel Media

Donna Strickland

Prize share: 1/4

The Nobel Prize in Physics 2018 was awarded "for groundbreaking inventions in the field of laser physics" with one half to Arthur Ashkin "for the optical tweezers and their application to biological systems", the other half jointly to Gérard Mourou and Donna Strickland "for their method of generating high-intensity, ultra-short optical pulses."

The Nobel Prize in Physics 2023



III. Niklas Elmehed © Nobel
Prize Outreach
Pierre Agostini
Prize share: 1/3

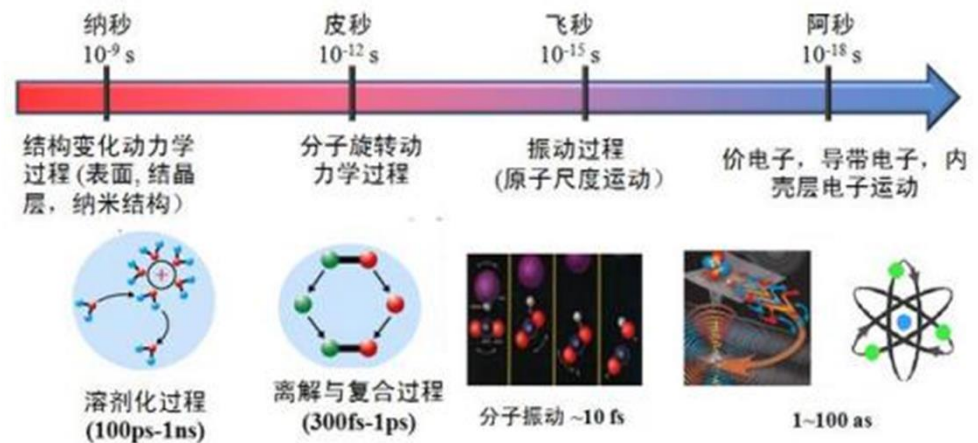


III. Niklas Elmehed © Nobel
Prize Outreach
Ferenc Krausz
Prize share: 1/3



III. Niklas Elmehed © Nobel
Prize Outreach
Anne L'Huillier
Prize share: 1/3

The Royal Swedish Academy of Sciences has decided to award the Nobel Prize in Physics 2023 to Pierre Agostini, Ferenc Krausz, and Anne L'Huillier "for experimental methods that generate attosecond pulses of light for the study of electron dynamics in matter".



Date	Name	Achievement
1900	Max Plank	Provided the understanding that light is a form of electromagnetic radiation
1916	Albert Einstein	Theory of light emission. Concept of stimulated emission.
1951	Charles H Townes	The inventor of the MASER - First device based on stimulated emission, awarded Nobel prize 1964.
1951	Alexander M. Prokhorov Nikolai G. Basov	Independent inventors of MASER at Lebedev Institute of Physics, Moscow. Awarded Nobel prize 1964.
1956	Nicolas Bloembergen	First proposal for a three-level solid state MASER at Harvard University.
1957	Charles H Townes	Sketches an early optical MASER in his lab book.
1957	Gordon Gould	First document defining a LASER; notarized by a candy store owner. Credited with patent rights in the 1970s.
1958	Arthur L Schawlow Charles H Townes	First detailed paper describing “Optical MASER”. Credited with invention of LASER. from Columbia University.
1959	Gordon Gould	Applies for LASER related patents

Date	Name	Achievement
1960	Arthur L Schawlow Charles H Townes	LASER patent No. 2,929,922.
1960	Theodore Maiman	Invented first working LASER based on Ruby. May 16th 1960, Hughes Research Laboratories.
1960	Ali Javan, William Bennett Donald Herriot	First helium-neon LASER at Bell Labs Dec. 1960, First gas laser and first CW laser.
1961	Leo F. Johnson, K. Nassau	First neodymium crystal LASER at Bell Labs
1962	Alan White Dane Rigden	First helium neon (HeNe) visible CW LASER at Bell Labs.
1964	Kumar N Patel	Inventor of CO ₂ LASER at Bell Labs.
1964	William Bridges	Invention of Argon Ion LASER at Hughes Labs.
1966	Peter Sorokin John Lankard	First dye LASER action demonstrated at IBM Labs.
1966	Mary L. Spaeth	First tunable dye LASER at Hughes Research Labs
1970	Nikolai Basov Yu M. Popov	First Excimer LASER at Lebedev Labs, Moscow based on Xenon (Xe) only.

Date	Name	Achievement
1972	Charles H. Henry	First quantum well LASER
1976	Jim Hsieh First	First InGaAsP diode LASER at MIT Lincoln Labs.
1976	John M J Madey's Group	First free electron LASER at Stanford University
1981	Arthur Schawlow Nicolas Bloembergen	Awarded Nobel Physics Prize for work in non-linear optics and spectroscopy
1982	Peter F. Moulton	First titanium sapphire LASER at MIT Lincoln Labs
1987	David Payne	First erbium fiber LASER amplifier
1994	Jerome Faist Federico Capasso Deborah L. Sivco Carlo Sirtori Albert Hutchinson Alfred Y. Cho	First quantum cascade multiple wavelength LASER at Bell Labs
1994	Nikolai Ledentsov	First quantum dot LASER at Ioffe Physico-Technical Institute.
1996	Wolfgang Ketterle	First pulsed atom LASER at MI

Date	Name	Achievement
1996		First Petawatt LASER at Lawrence Livermore National Labs.
1997	Wolfgang Ketterle	First atom LASER at MIT Lincoln Labs.
2004	Ozdal Boyraz Bahrom Jalali	First silicon Raman LASER at the University of California, Los Angeles
2006	John Bowers	First silicon LASER
2007	John Bowers Brian Koch	First mode-locked silicon evanescent LASER
2010		First 10 Petawatt LASER at Lawrence Livermore National Labs.